JÖNKÖPING UNIVERSITY
School of Engineering

Networks and the Internet

Lecture 4, Information Security, 7,5 ECTS

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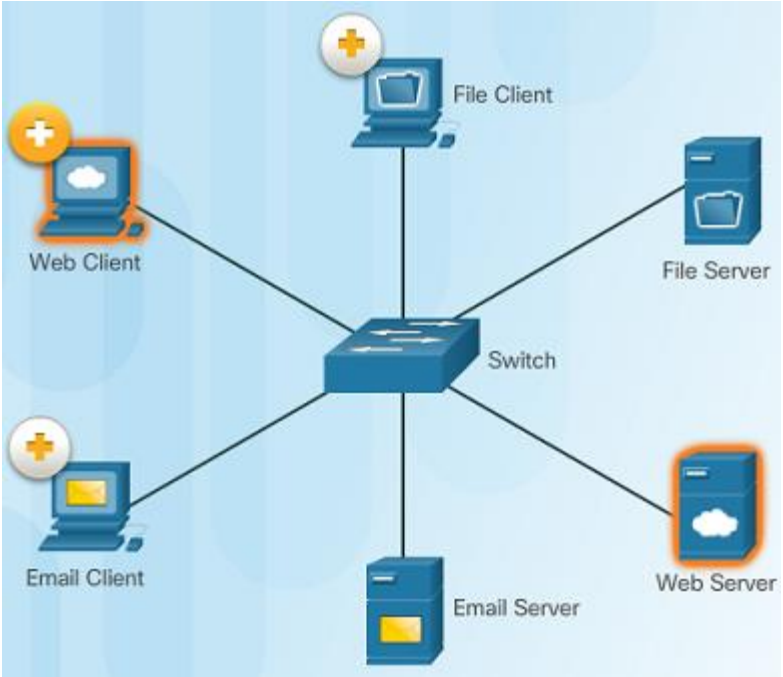
Objectives

- Resources within a network
- Protocols and standards
- Models
- Internet history
- IP and MAC addresses

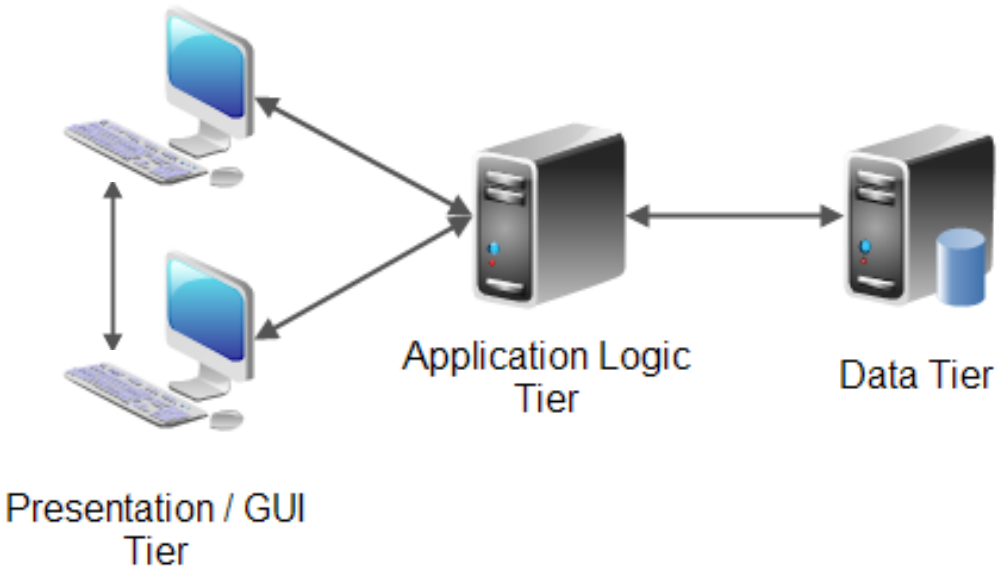
The evolution of communication

Resources in a network

- Every computer connected to a network is called a host or end device
- Servers are computers that provide information to end devices on the network, e.g., email servers, web servers, or file servers
- Clients are computers that request information from the servers



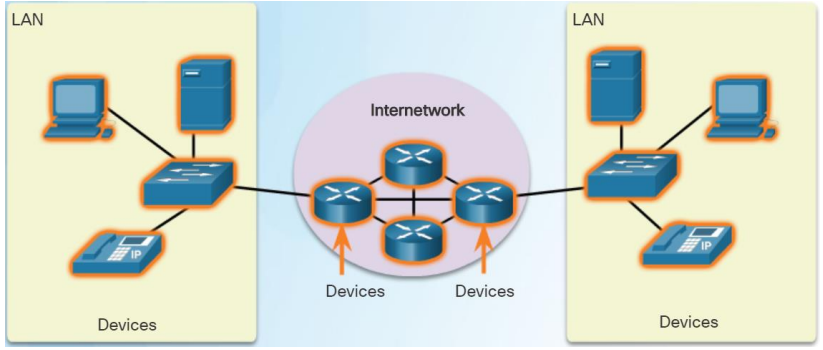
How are resources provided?

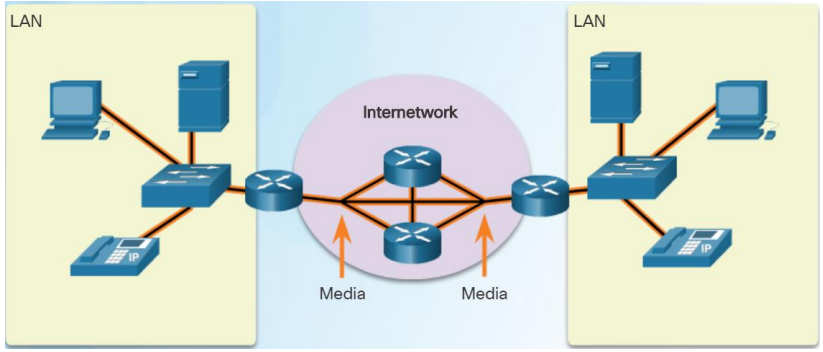


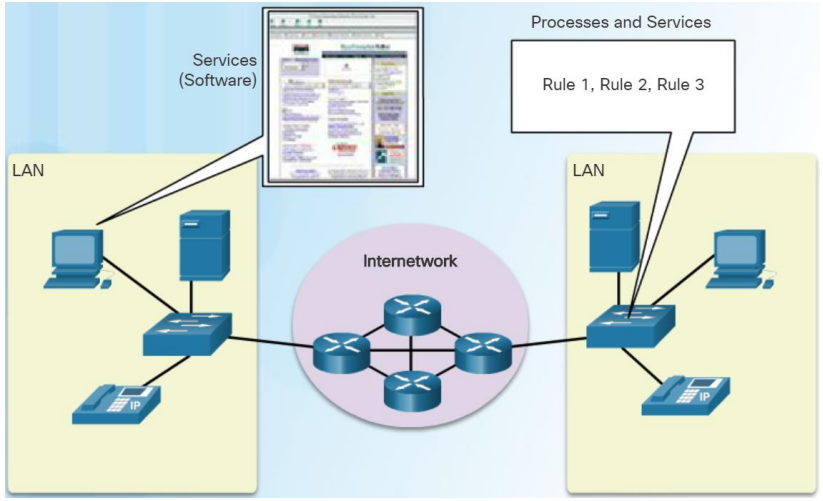
- Traditional client/server (architecture)
  - Two-tier
- Multi-tier architecture
  - Normally three-tier, but can be n-tier
- Example three-tier
  - Presentation tier (web browser)
  - Application tier (webserver)
  - Data-tier (database server)
- Peer-to-peer (P2P) architecture
  - Two-tier (ex. is filesharing, torrents)

Components of a network

- Devices
  - End devices
    - Source or destination of a message
  - Intermediary network devices
    - Routers, switches, firewalls,…
    - Regeneration of signals, error notification, prioritization,…
- Media
  - Copper – data encoded into electric impulses
  - Fiber-optic – data encoded as pulses of light
  - Wireless – data encoded using wavelengths from the electromagnetic spectrum
- Services
  - Implicit - DHCP, DNS,…
  - Explicit – email, web hosting,….

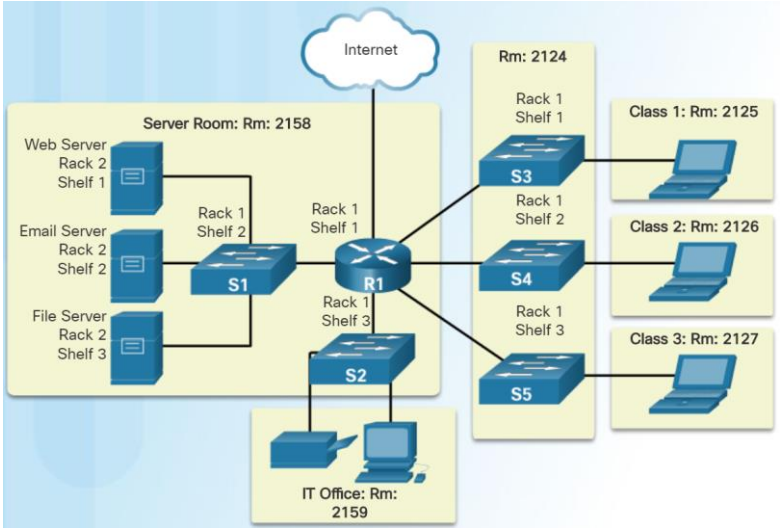




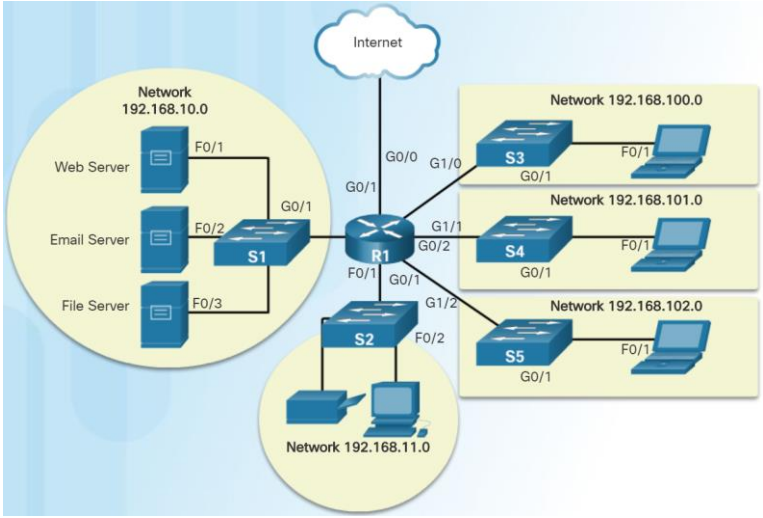


Topology diagrams

- Physical topology
  - Identify the physical location of intermediary devices and cable installation



- Logical topology
  - Identify devices, ports, and addressing scheme

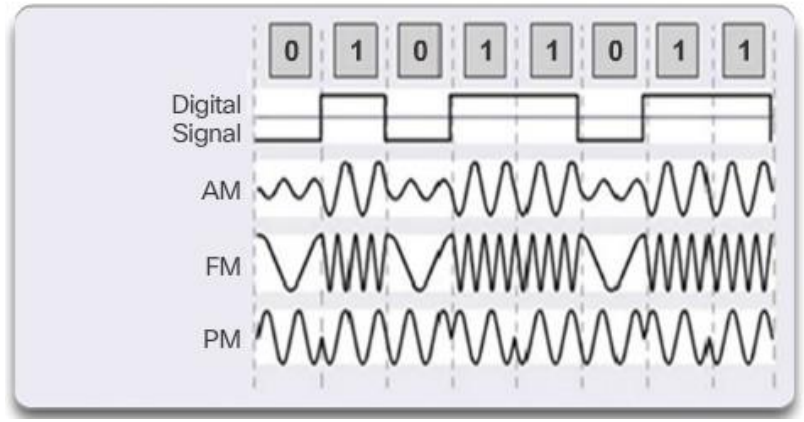


Physical layer

- The physical connection to a local network
- We need the following physical layer elements
  - The physical media and associated connectors
  - A representation of bits on the media
  - Encoding of data and control information
  - Transmitter and receiver circuitry on the network devices
- Three basic forms of network media
  - Copper cable - electrical
  - Fiber - optical
  - Wireless - microwave

Electrical Signals - Copper cable

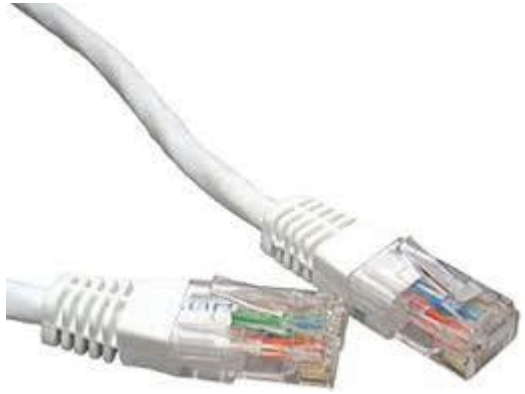
Light Pulse - Fiber-optic cable

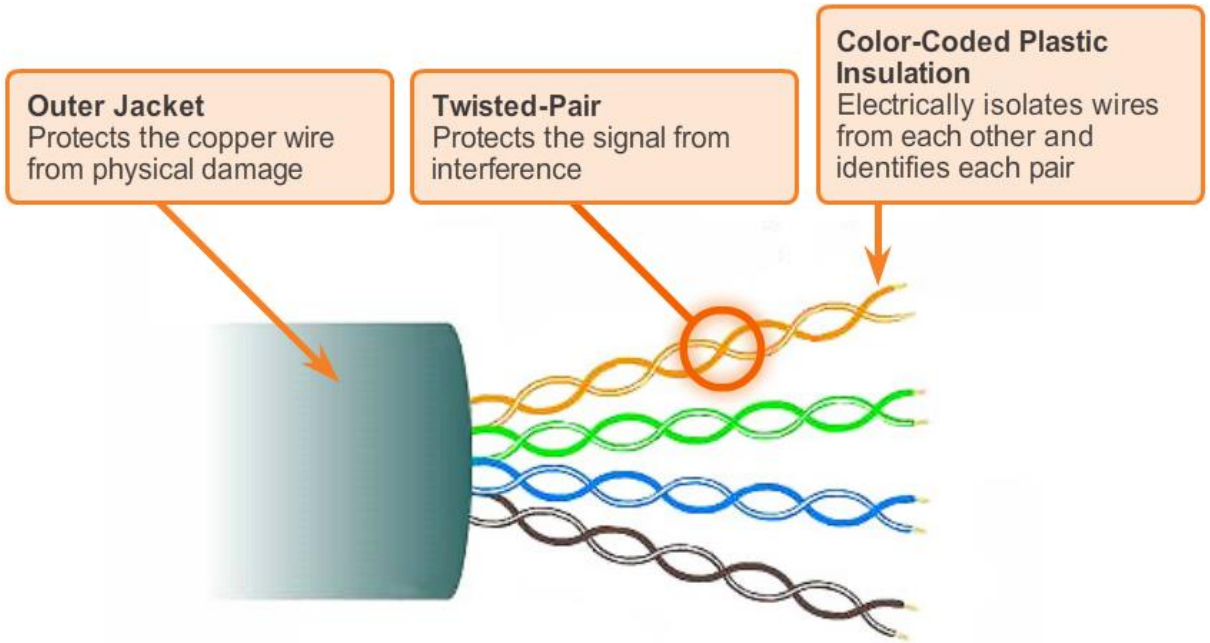


Microwave Signals - Wireless

Unshielded Twisted-Pair (UTP)

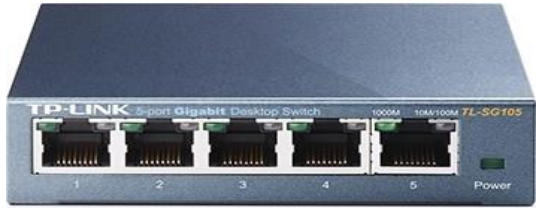
- UTP cabling is the most common networking media
  - Terminated with RJ-45 connectors
  - Used for interconnecting network hosts with networking devices, e.g., switches
  - It consists of four pairs of color-coded wires that have been twisted together to help protect against signal interference from other wires
  - Color codes aid in cable termination



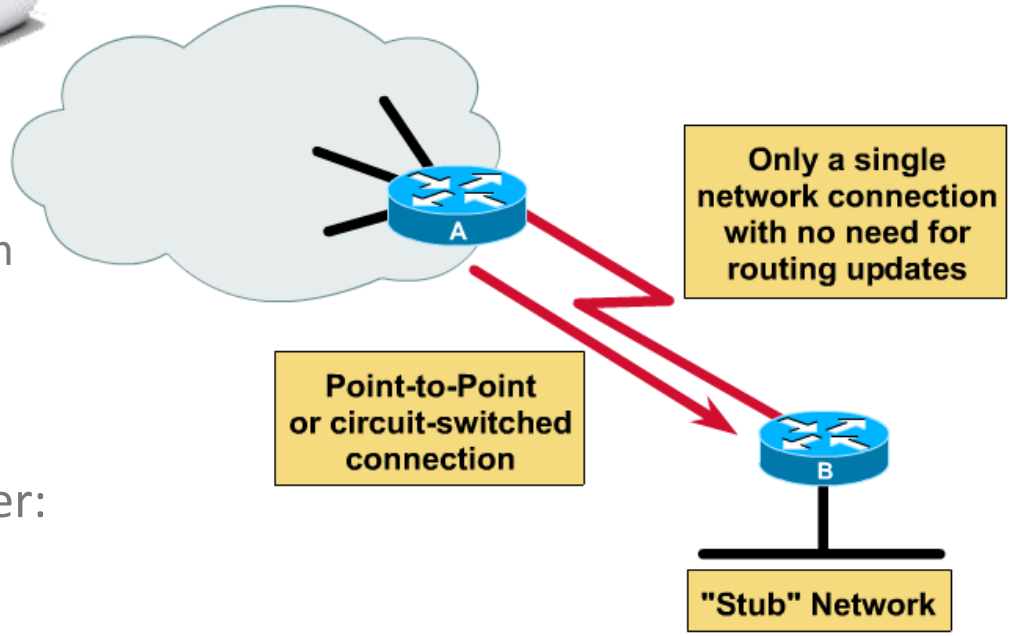


Physical connection

- The Hub (hub is a multiport repeater)
  - The hub is the simplest connection device, can't direct packets
  - A simple point of distribution
- The Repeater
  - Take a weakened signal and boosts it
  - Regenerates the signal, to avoid resending noise
- The Switch
  - A smart Hub that can direct packets
  - Full-Duplex Operation
  - Dedicated Bandwidth and multiple ports N *(N is often 6, 8, 16, 24 or 48/…)*
  - Port bit rate is variable (often 10/100/1000/… Mbps)

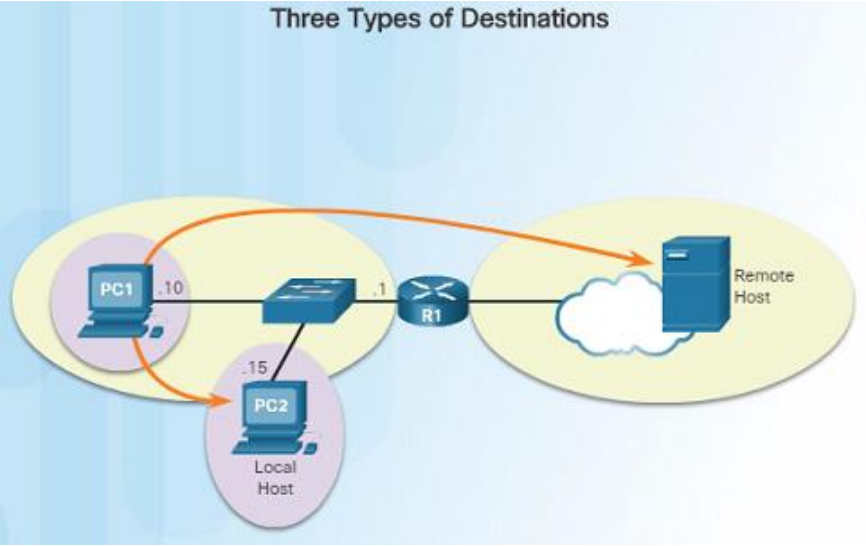


Routing types



- Routers are:
  - Directly connected to networks (usually via cable with IP on an interface)
  - Non-directly connected to networks (remote networks connected to other routers)
- A router learns about non-directly connected networks either:
  - Static
    - Uses a programmed route that an administrator enter
  - Dynamic
    - Uses a route that a routing protocol adjusts automatically for topology or traffic changes (e.g., distance vector routing protocols and link-state routing protocol)

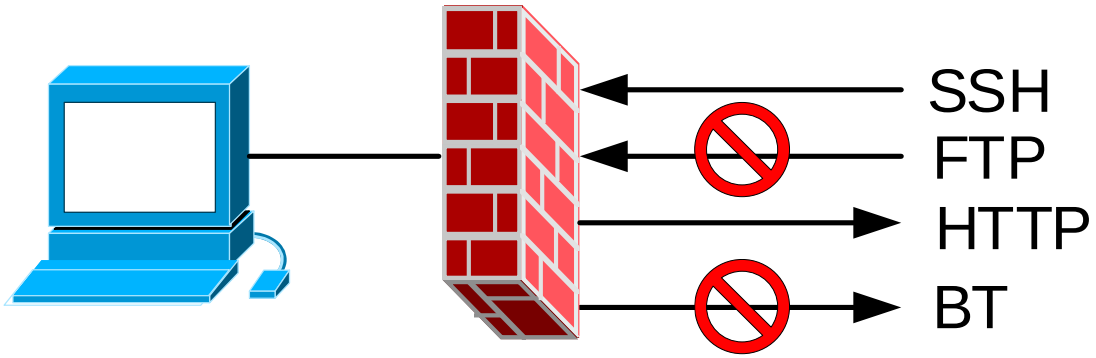
Routing and routing operation



- A host could send to:
  - Itself, 127.0.0.1, direct connection
  - Localhost, same network
  - Remote host (other network)
- When a package arrives, the destination IP address is extracted.
- The router checks the routing table for a match and:
  - If it finds a match, the packet is forwarded to the interface and sent
  - If it doesn't find a match but has a default route, it sends it to the default route
  - If it doesn't find a match and has no default route, it drops the packet

Firewalls

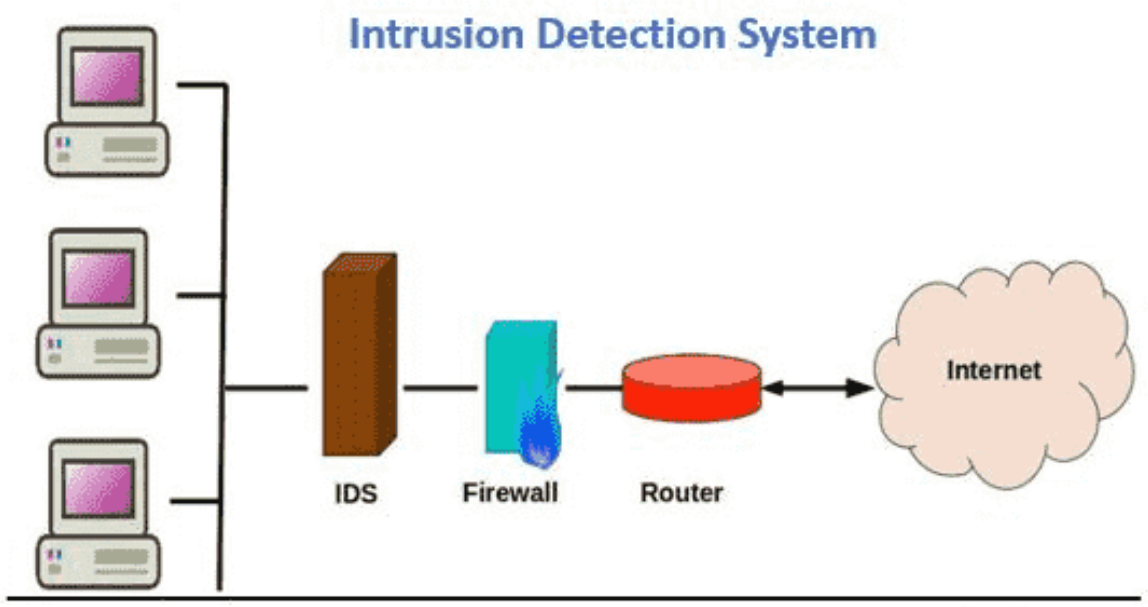
- A device that filters all traffic between a protected or “inside” network and less trustworthy or “outside” network
- Most firewalls run as dedicated devices
  - Easier to design correctly and inspect for bugs
  - Easier to optimize for performance
- Firewalls implement security policies or rule-sets that determine what traffic can or cannot pass through
- A firewall is an example of a reference monitor, i.e., three characteristics:
  - Always invoked (cannot be circumvented)
  - Tamperproof
  - Small and simple enough for rigorous analysis



Intrusion detection and prevention

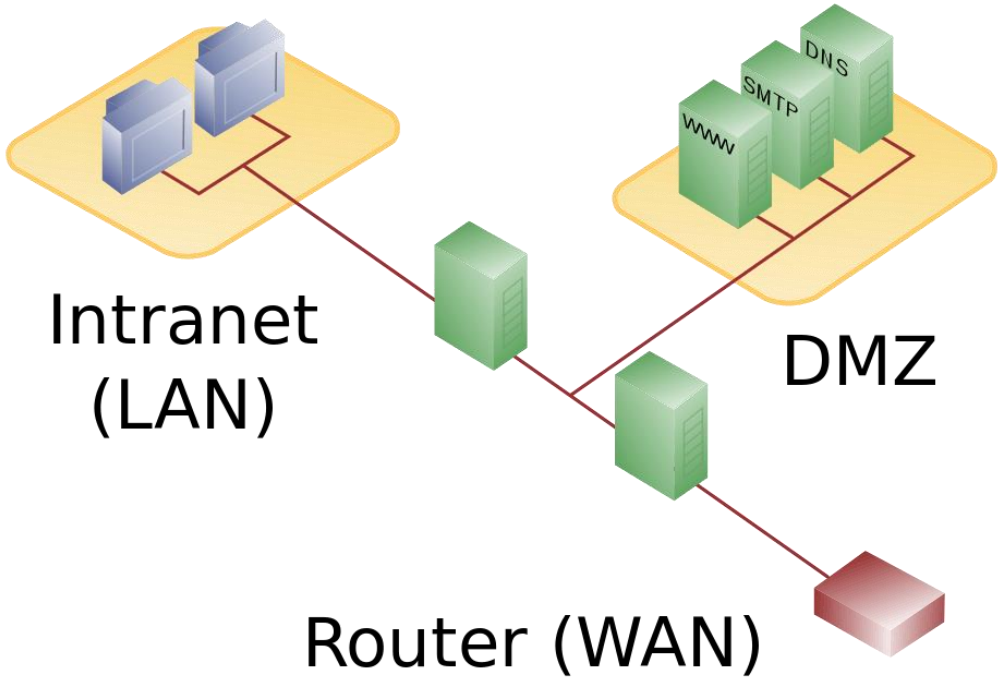
- Intrusion Detection Systems (IDS) and Intrusion Prevention Systems (IPS) complement other controls
- IDSs monitor activity to identify malicious or suspicious events

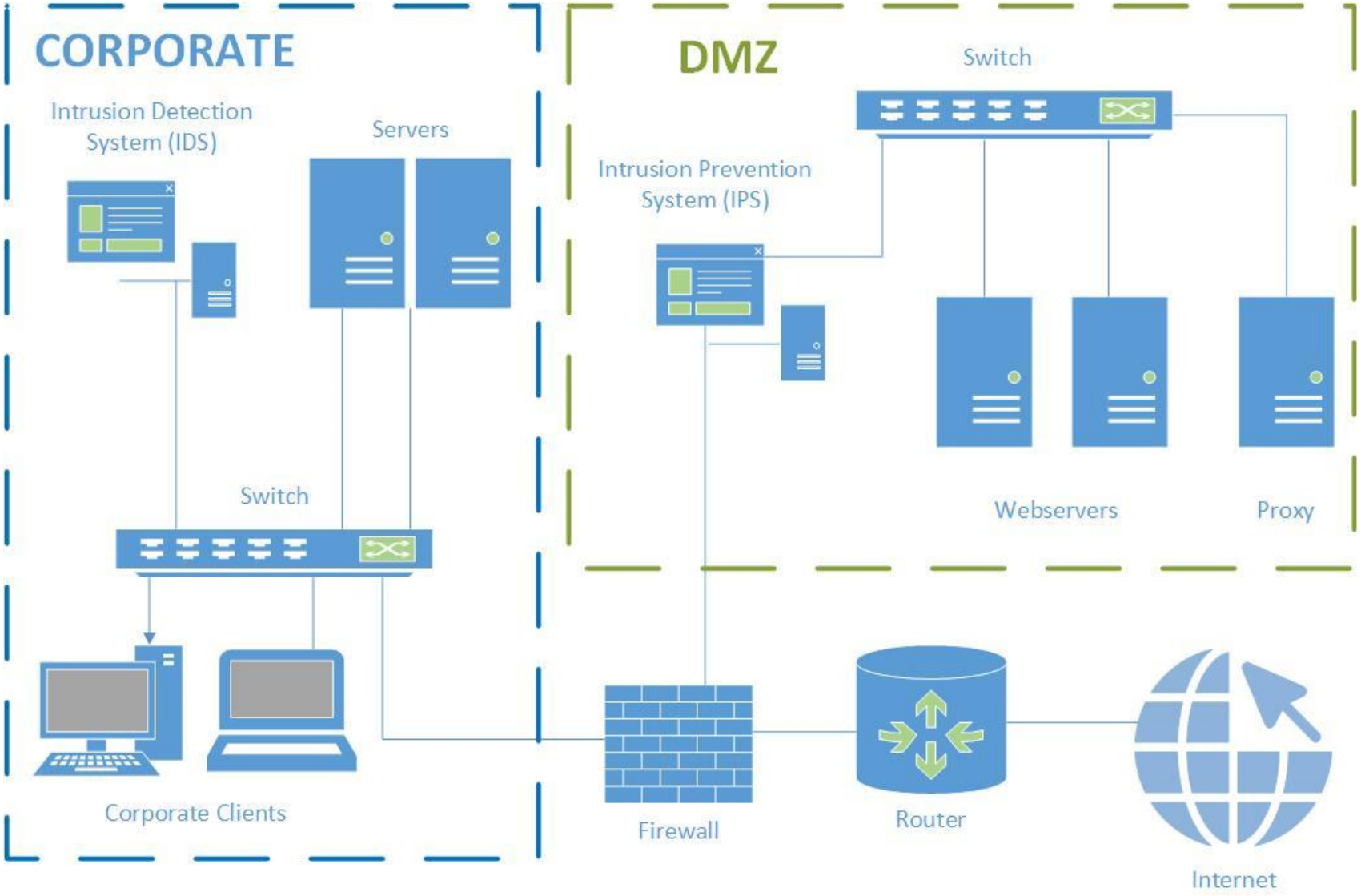
- IDSs may:
  - Monitor user and system activity
  - Audit system configurations for vulnerabilities and misconfigurations
  - Assess the integrity of critical system and data files
  - Recognize known attack patterns in system activity
  - Identify abnormal activity through statistical analysis
  - Manage audit trails and highlight policy violations
  - Install and operate traps to record information about intruders



Demilitarized zone

- DMZ, perimeter network or screened subnet
- Physical or logical subnetwork
- DMZ is a form of network architecture
- Services dedicated to outside use separated
- The idea is that intrusion of DMZ hosts lead to only limited damage to the internal hosts



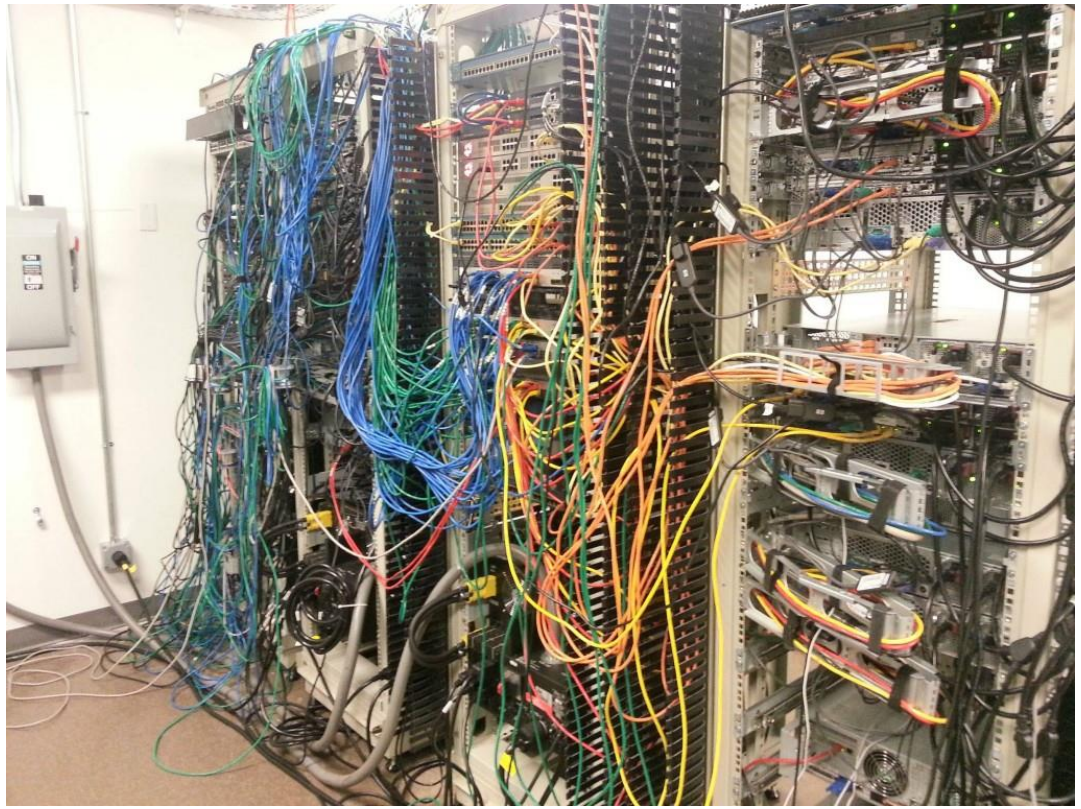

CORPORATE
Intrusion Detection System (IDS)
Servers
Switch
Corporate Clients
DMZ
Switch
Intrusion Prevention System (IPS)
Webservers
Proxy
Firewall
Router
Internet

Interruption

- Routing
  - Routing protocols are complicated, and one misconfiguration or deliberate attack can poison the data of many routers
- Excessive demand
  - Network capacity is finite and can be exhausted; an attacker can generate enough demand to overwhelm a critical part of a network
  - Like DoS/DDoS (soon…)
- Component failure
  - Component failures tend to be sporadic and unpredictable and will cause loss of service if not planned for

Keeping information safe

- Networks are central to our society!
  - Communication
  - Transactions…
- We depend on networks for ”everything”
  - It’s even hard to avoid networking
- What is the network?
  - Where are the borders?
  - Complex structures…
- Ok, networks are important. I get it!
  - That’s why we need to protect them!
  - Hence, the design, development, and usage of networks are critical!



Wireless network security

- We still need to preserve CIA
- Wireless security is similar to a wired environment in many ways
  - But the underlying communications medium is the most significant source of risk
- Wireless could be many things
  - Here wireless = WLAN = 802.11 = Wi-Fi
- Wi-Fi used to be insecure… now it's much better
- Vulnerabilities include:
  - The media itself
    - Prone to eavesdropping
    - Shared media = easy insertion and easy disruption (DoS)
  - Who do you connect to?
  - Protocols (such as WEP (and WPA))

Types of wireless media

- Wireless Personal Area Network (WPAN) - IEEE 802.15
  - Bluetooth, Zigbee, Z-wave, WUSB (Wireless Universal Serial Bus)
  - Range: 1-10-100m
  - Speed: 2Mbps(Bluetooth)-480Mbps(WUSB)
- Wireless Local Area Network (WLAN) - IEEE 802.11
  - IEEE 802.11a/b/g/n/ac/ad/…
  - Range: 10-100m (indoor), 100-5000m (outdoor)
  - Speed: 1Mbps-7Gbps
- Wireless Metropolitan Area Network (WMAN) - IEEE 802.16
  - WiMAX (Worldwide Interoperability for Microwave Access)
  - Range: up to 10km
  - Speed: up to 70Mbps
- Other Technologies include e.g.
  - Radio-frequency identification (RFID) – To identify and track the data of things
  - Sensors - to detect events or changes in its environment and send the information to e.g. things

Wireless network security

- On a higher level, we have developed:
  - WEP – Wired Equivalent Privacy (bad, don’t use it)
  - WPA- Wi-Fi Protected Access (better)
  - WPA2/802.11i (best, and standard (on hardware) since 2004-2006)
    - Still not perfect, e.g., vulnerable to DoS
  - WPA3 since 2018, 192-bit cryptographic strength in WPA3-Enterprise mode
    - WPA3 support has been mandatory for devices which bear the "Wi-Fi CERTIFIED™" logo since July 2020
- WPA, WPA2 and WPA 3 can be either:
  - Personal
    - Using a pre-shared key, No authentication server, home and small office use, a shared key is a security issue (e.g., guessing attacks)
  - Enterprise
    - Uses authentication server, no pre-shared key, bigger organisations, use IEEE 802.1X (so the access point become the authenticator), mutual authentication (i.e., no man-in-the-middle)

International networking standards organisations

- ISO (International Organization for Standardization)
  - OSI-model, ISO/IEC 8859 (8-bit single-byte coded graphic character sets),…
- IEEE (Institute of Electrical and Electronics Engineers)
  - IEEE 802 – (LAN/MAN), for example, 802.11 (WLAN),…
- IETF (Internet Engineering Task Force)
  - TCP/IP,…
- ITU-T (The International Telecommunications Union-Telecommunication)
  - IPTV, DSL,…
- TIA (Telecommunications Industry Association)
  - VoIP devices,…
- EIA (Electronic Industries Alliance)
  - Connectors, wiring,…
- ICANN (The Internet Corporation for Assigned Names and Numbers)
  - UDP and TCP port numbers,…
- …

Example: IEEE

- 802 has LAN/MAN, including:
  - 802.1 Higher Layer LAN Protocols
  - 802.3 Ethernet
  - 802.11 Wireless LAN
  - 802.15 Wireless Personal Area Network (WPAN)
  - 802.16 Broadband Wireless Access
  - 802.19 Wireless Coexistence
  - 802.20 Wireless Mobility
  - 802.21 Media Independent Handover Services
  - 802.22 Wireless Regional Area Networks
  - 802.24 Smart Grid

http://grouper.ieee.org/

Workgroup P802

Ethernet

- Ethernet is the most widely used LAN technology today
- Ethernet specifications support different media, bandwidths, and other Layer 1 and 2 variations
- The basic frame format and address scheme is the same for all varieties of Ethernet
- It has evolved from a shared media, contention-based, half-duplex data communications technology to high bandwidth, full-duplex technology
- The LLC handles the communication between the upper layers and the lower layers
  - Implemented in software and independent of the hardware.
- The MAC sublayer has two primary responsibilities:
  - Data encapsulation
    - Frame delimiting
    - Addressing
    - Error detection
  - Media access control
    - Responsible for the placement and removal of frames

Communication fundamentals

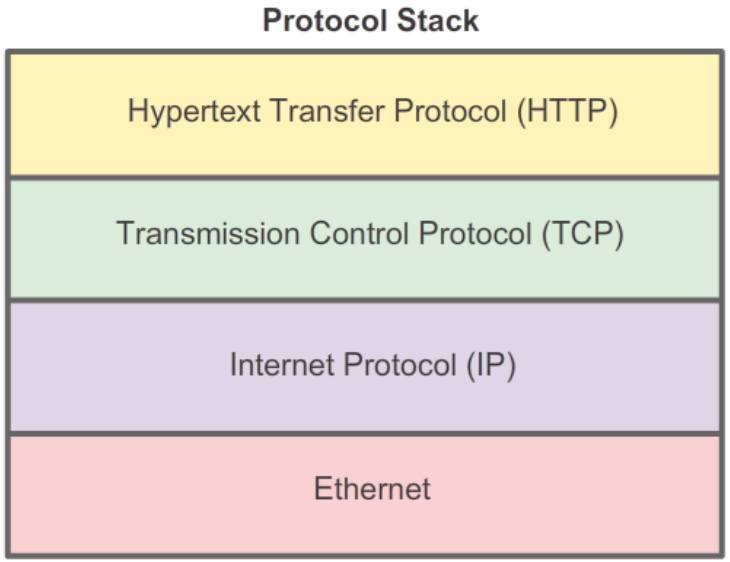
- Before beginning to communicate with each other, we establish rules or agreements to govern the conversation
- These rules, or protocols, must be followed in order for the message to be successfully delivered and understood
- A protocol is nothing more than an agreement or rules to govern a way of communicating
- Among the protocols that govern successful human communication are:
  - An identified sender and receiver
  - Agreed upon method of communicating (face-to-face, telephone,…)
  - Common language and grammar
  - Speed and timing of delivery
  - Confirmation or acknowledgment requirements

Rules of communication

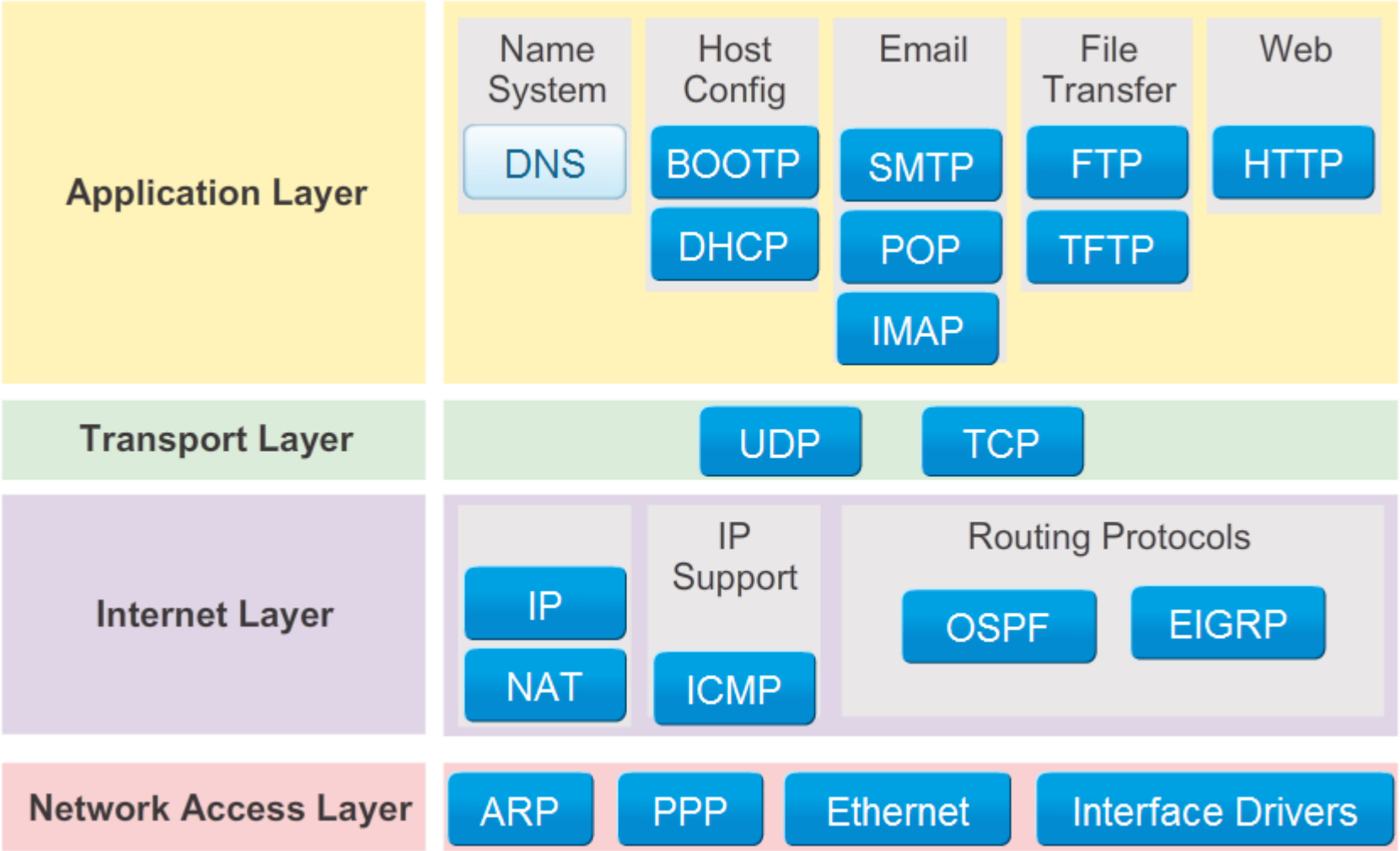
- Protocols are necessary for effective communication and include:
  - Identified sender and receiver
  - Common language and grammar
  - Speed and timing of delivery
  - Confirmation or acknowledgment requirements
- Protocols used in network communications also define:
  - Message Encoding
    - Process of converting information into another acceptable form
  - Message Formatting and Encapsulation
    - Agreed format for proper delivery and encapsulated into a frame
  - Message Size
    - Long messages must be broken into smaller pieces (packets)
  - Message Timing
    - Access method – when do we send?
    - Flow control – How much do we send?
    - Response timeout – How long do we wait?
  - Message Delivery Options
    - Unicast, multicast, broadcast

Protocols

- Protocols are often designed for a single purpose
  - Modularization to a stack makes design easier
- Services are associated with protocols, for example:
  - Service WWW – Protocol HTTP
  - Service e-mail – Protocol SMTP,POP,IMAP
  - Service IP-telephony – Protocol SIP, H323
- The protocol suite is the definition of the protocols
- The protocol stack is the software implementation



TCP/IP protocol suite

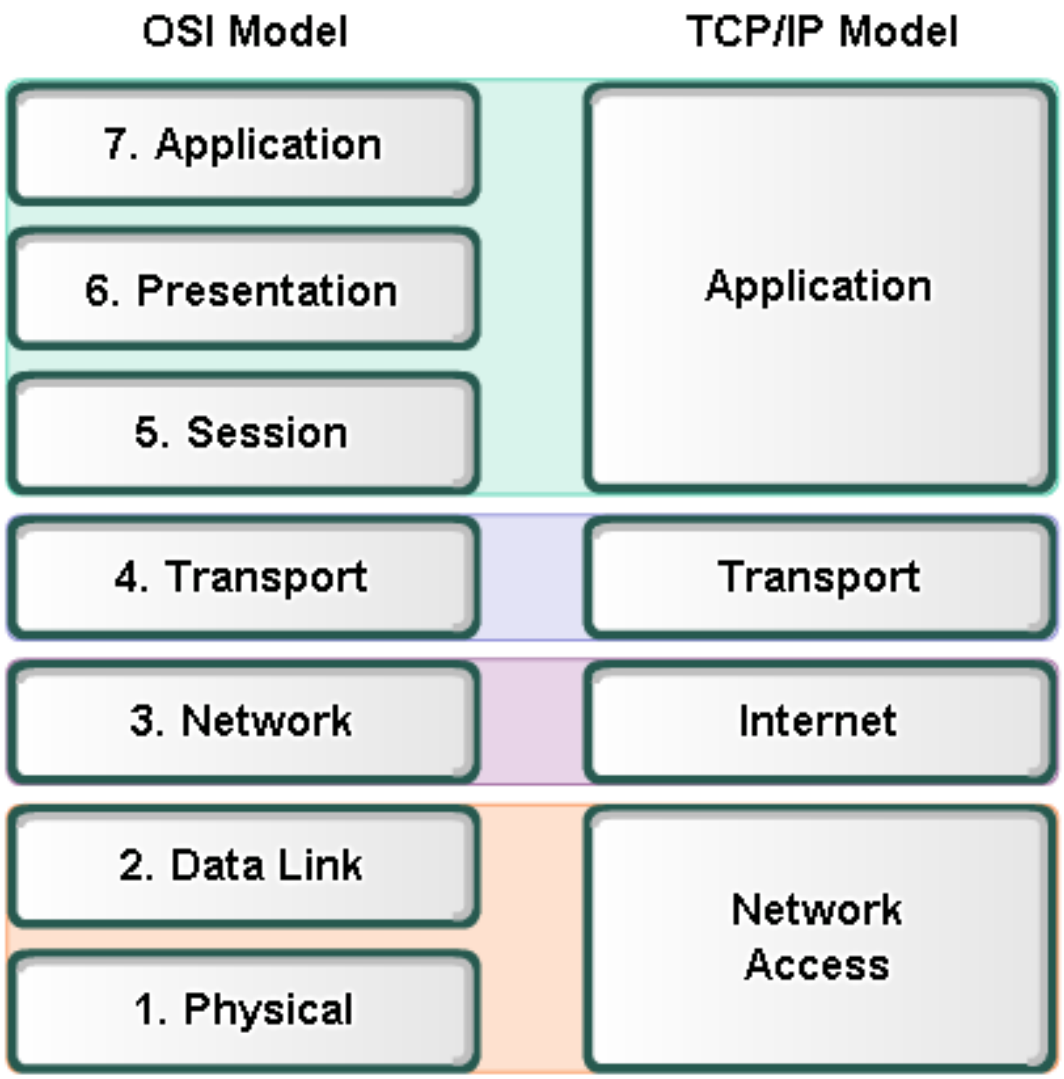


Layered models

- There are benefits to using a layered model to describe network protocols and operations:
  - Assists in protocol design
  - Fosters competition because products from different vendors can work together
  - Prevents technology or capability changes in one layer from affecting other layers above and below
  - Provides a common language to describe networking functions and capabilities

The models

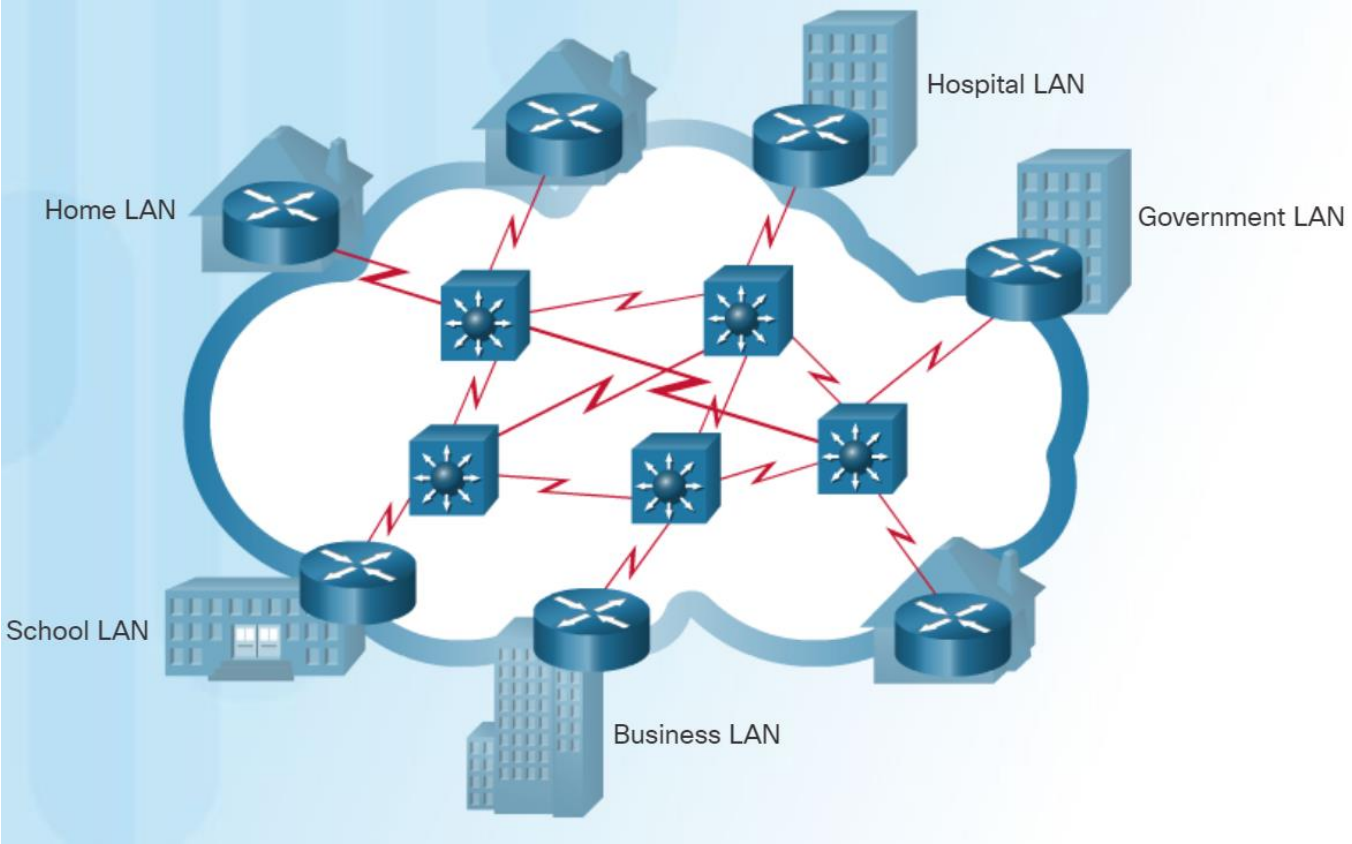
- More accurately models actual implementations of modern communication subsystems
- Name derived from two of the core protocols
  - TCP: Transmission Control Protocol
  - IP: Internet Protocol
- Now used for other applications/protocols as well
  - Required modification of network interface layers to be applicable for other network types
- A five-layer version also exists, where:
  - Network Access is divided into the Data-link layer and Physical layer

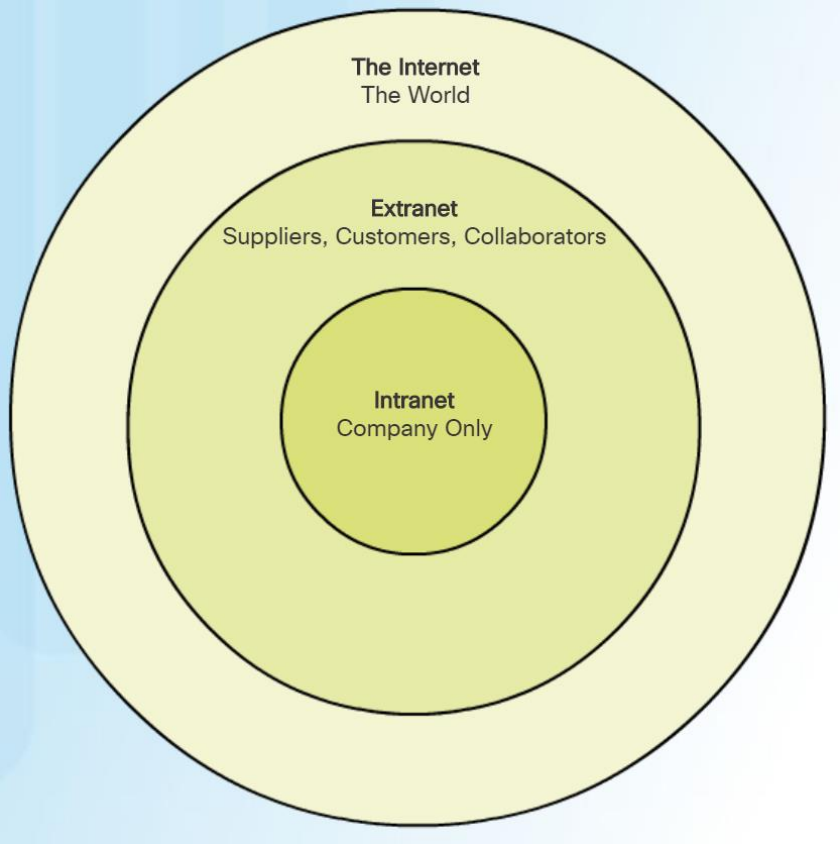


History of the Internet

- Started as an ARPA (Advanced Research Projects Agency) project
  - The aim was to connect several physical networks into one logical
  - The first ARPANET link was between UCLA and Stanford in 1969 (using ideas from ALOHAnet)
  - Development was centered around RFCs (Request For Comments)
  - 1971 the first e-mail was sent
  - The first Swedish link was via satellite in 1973
  - Work with TCP/IP began in 1974
  - The word Internet was first used in an RFC in 1974
- Parallel development for creating a worldwide business network, X.25 started in 1975 (mainly in Europe)
- In 1983 TCP/IP became the only approved ARPANET protocol
- In the late ‘80s ISPs started to pop up, opening the network for commercial uses
- In 1991 Tim Berners-Lee (CERN) launched the WWW idea
- In 1993 the Mosaic web browser was invented by Mark Andreesen
- …

Internet/Extranet/Intranet

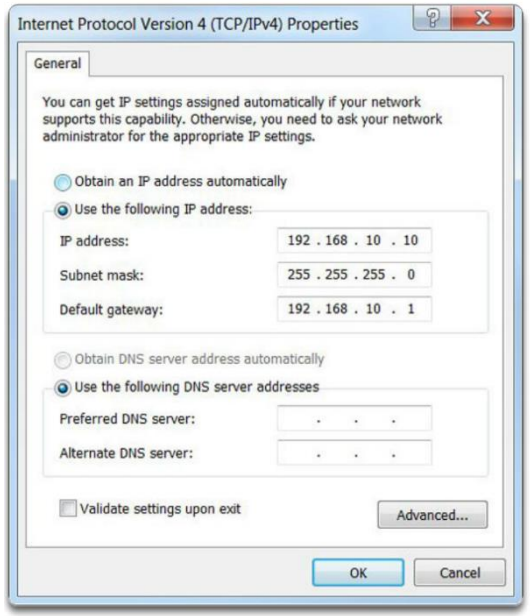




IPv4 addresses

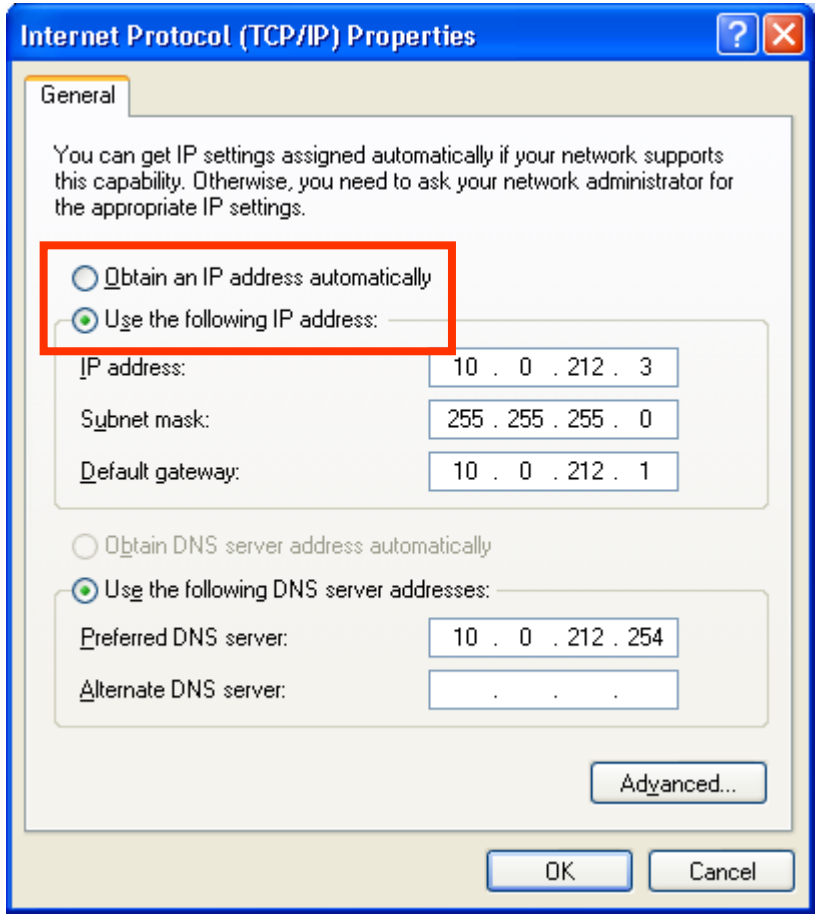
- The IP address is a hierarchical address that is made up of two parts:
  - a network portion and
  - a host portion
- When a host is configured, the IP address and subnet mask are assigned.

	Network Portion			Host Portion
IPv4 Address	192 .	168 .	10 .	10
	11000000	10101000	00001010	00001010
Subnet Mask	255 .	255 .	255 .	0
	11111111	11111111	11111111	00000000



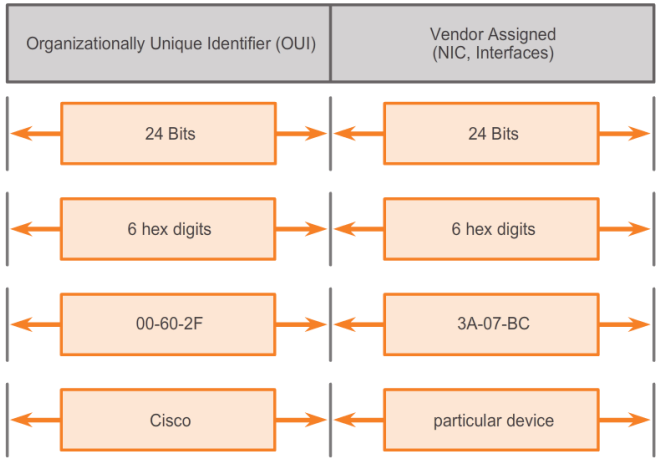
IPv4 addresses for users

- Static Assignment
  - Useful for printers, servers, networking equipment, and other “central” devices…
  - Keep to a minimum!
- Dynamic Assignment
  - DHCP (Dynamic Host Configuration Protocol)



MAC addresses (also called Physical)

- Part of the Ethernet protocol includes the MAC
- Every Ethernet NIC card has a unique MAC address
- MAC addresses provide a way for computers to identify themselves
- They give hosts a permanent, unique name
- MAC addresses are:
  - 48 bits in length
  - Expressed as 12 hexadecimal digits
    - The first 6 are administered by the IEEE, and identify the manufacturer or vendor, called OUI (Organizational Unique Identifier)
    - The last 6 comprise the interface serial number or another value administered by the specific vendor
  - Sometimes called BIA (Burned-In Addresses) because they are burned into ROM
  - They are then copied into RAM when the NIC initialises
  - Sometimes called physical address
- MAC for local use (LAN) and IP for global use (WAN)
- What is my MAC address? Use command prompt: **ipconfig /all**



With Dashes 00-60-2F-3A-07-BC

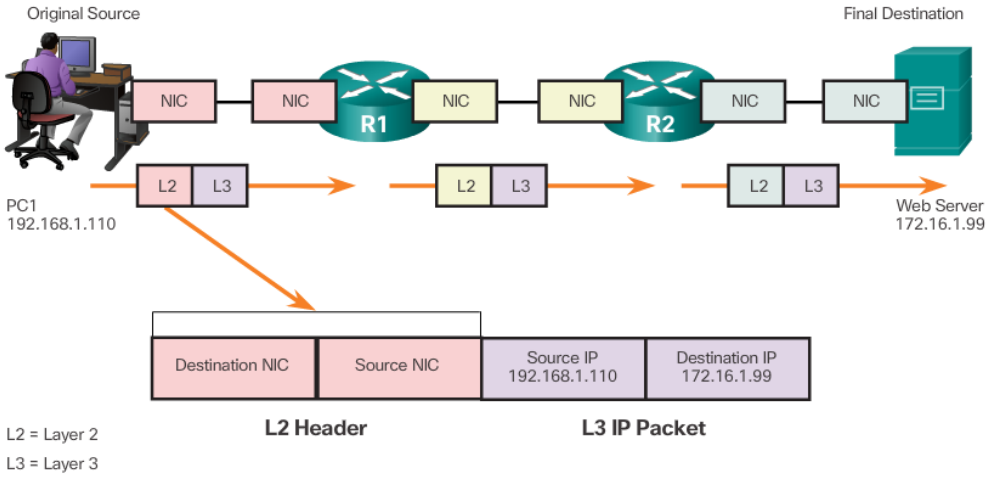
With Colons 00:60:2F:3A:07:BC

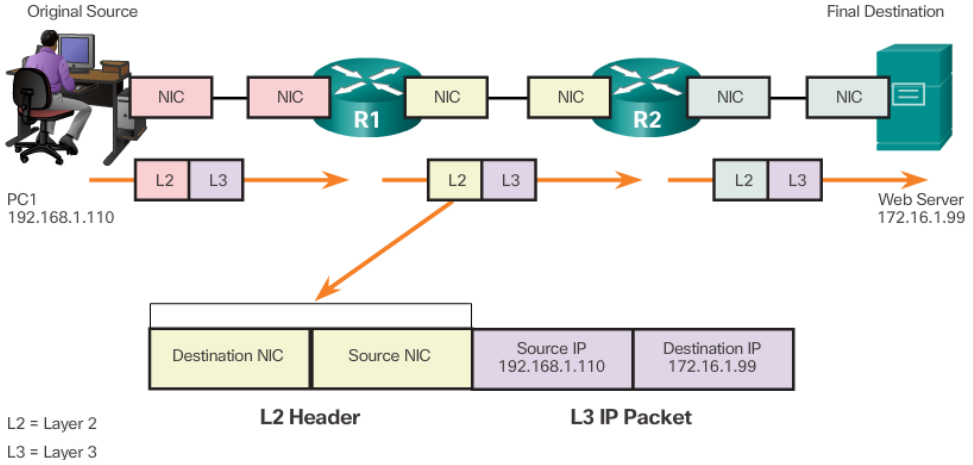
With Periods 0060.2F3A.07BC

The combination of MAC and IP facilitates End-to-End communication.

IP and MAC

- Destination on the same network
  - Physical address (MAC address) is used for Ethernet Network Interface Card (NIC) to Ethernet NIC communications on the same network.
- Destination on a remote network
  - A logical address (Internet Protocol (IP) address) sends the packet from the original source to the final destination.



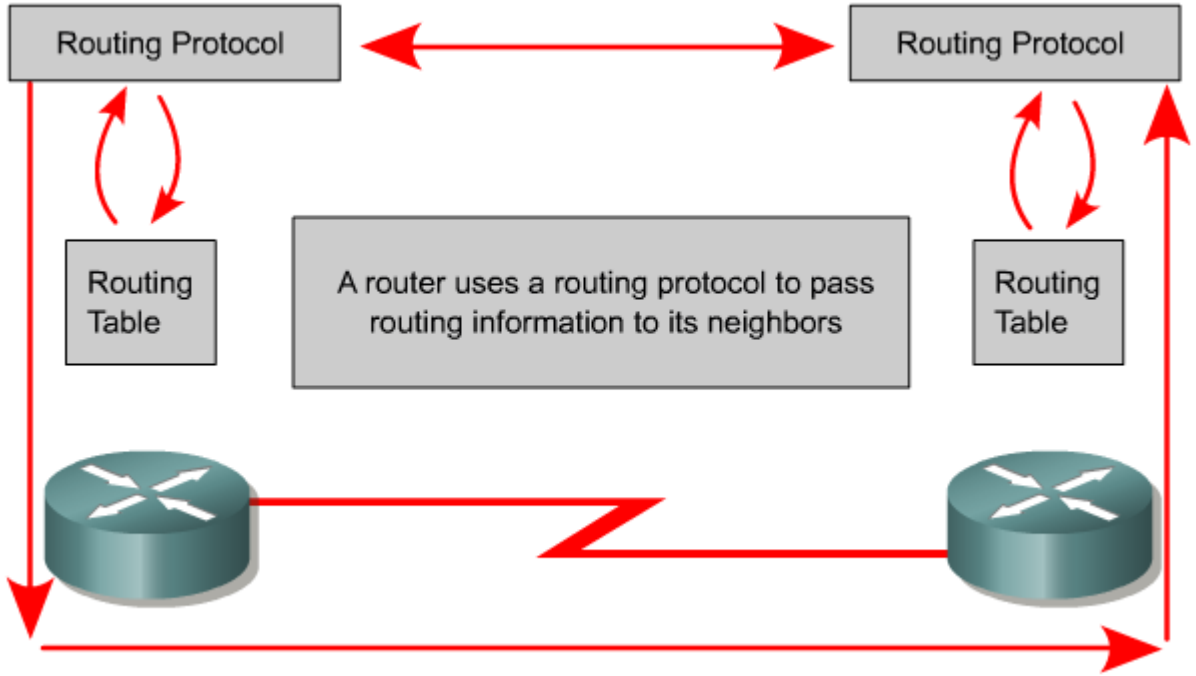


Address Resolution Protocol

- ARP (Address Resolution Protocol) is a Network Layer protocol that:
  - It is included in the TCP/IP protocol suite
  - Resolves IPv4 addresses to MAC addresses (L3 to L2)
  - Maintains a cache of mappings called ARP table
    - The table is stored in the RAM
    - Each row in the table contains one IP and one MAC address
    - Timers are used to keep up-to-date content
- A host can collect ARP table entries by:
  - Collecting from traffic by monitoring the network
  - Broadcast ARP requests
- Removing Entries from an ARP Table
  - Entries are removed from the device's ARP table when its cache timer expires.
  - Cache timers are OS-dependent.
  - ARP entries can be manually removed via commands.
- Problems are mainly:
  - Overhead
  - Security problems such as ARP poisoning (ARP spoofing)

Routing protocols

- The goal of a routing protocol is to build and maintain the routing table
- This table contains the learned networks and associated ports for those networks
- Routers use routing protocols to manage information received from:
  - other routers
  - information learned from the configuration of its interfaces
  - manually configured routes



What's your address?
173.168.15.10
VIA 9GAG.COM
No, your local address.
127.0.0.1
I mean your physical address.
29:01:38:62:31:58

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